

University of Plymouth

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PROJ519: MSc Dissertation and Research Skills

Project Proposal

A SHAP-Based Explainability and Fairness Audit Framework for Fraud Detection Models in UK Fintech: A Consumer Duty Compliance Perspective

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1. Introduction to the Research Area

UK financial technology (fintech) firms increasingly rely on machine learning (ML) models to detect fraudulent transactions in real time, replacing or augmenting rule-based systems that struggle to keep pace with evolving fraud typologies. Gradient-boosted tree ensembles and deep learning classifiers now routinely outperform traditional logistic regression approaches in fraud detection benchmarks, but this performance gain comes at the cost of interpretability: the internal decision logic of these models is largely opaque to the analysts, compliance officers, and customers affected by their outputs.

This opacity has become a regulatory concern. The Financial Conduct Authority's (FCA) Consumer Duty, which came into full force in 2023, requires firms to act to deliver good outcomes for retail customers, including the avoidance of foreseeable harm and the provision of clear, understandable communication about decisions that affect them (Financial Conduct Authority, 2022). A fraud detection model that blocks a legitimate transaction, freezes an account, or flags a customer for additional checks is making a consequential decision about that customer. Where such decisions cannot be explained, firms are exposed to both conduct risk and the reputational and legal consequences of an inability to justify automated outcomes.

Explainable AI (XAI) techniques, and SHapley Additive exPlanations (SHAP) in particular, have emerged as a leading approach for attributing model predictions to individual input features in a theoretically grounded, model-agnostic way (Lundberg and Lee, 2017). A separate but related concern is algorithmic fairness: fraud models trained on historical transaction data can inadvertently learn proxies for protected characteristics such as age, income bracket, or geographic location, producing disparate false-positive rates across customer subgroups (Mehrabi *et al.*, 2022). For a fintech firm, this raises the question of whether a model that performs well in aggregate is nonetheless treating some customer groups unfairly, an outcome Consumer Duty would also classify as harm.

This dissertation will examine SHAP-based explainability and fairness auditing as a combined framework for evaluating fraud detection models and will assess the extent to which such a framework can support the kind of evidence base a UK fintech firm would need to demonstrate Consumer Duty compliance.

2. Aims and Objectives

Aim: To develop and evaluate a combined explainability and fairness audit framework for machine learning-based fraud detection models, and to assess its suitability as a tool for demonstrating compliance with the FCA Consumer Duty.

Objectives:

- To critically review the literature on explainable AI methods, fairness metrics, and UK financial regulation as applied to automated decision-making in financial services.
- To design, train and validate a representative fraud detection model using a public transaction dataset, benchmarking at least two model architectures (e.g. logistic regression as an interpretable baseline, and a gradient-boosted ensemble such as XGBoost or LightGBM).
- To apply SHAP to generate global and local explanations of model behaviour, identifying the features that most strongly drive fraud predictions.

- To quantify fairness across relevant customer subgroups using established group fairness metrics (e.g. demographic parity, equalised odds, disparate impact ratio), using available demographic or proxy variables in the dataset.
- To evaluate the trade-offs observed between predictive performance, explainability and fairness, and to map these findings against the specific obligations of the Consumer Duty (Consumer Understanding and Avoidance of Foreseeable Harm outcomes).
- To produce practical recommendations for how UK fintech firms could operationalise explainability and fairness auditing within existing model risk management processes.

3. Research Context: Summary Review of Relevant Literature

Three bodies of literature underpin this project: explainable AI methods, algorithmic fairness, and the regulatory treatment of automated decision-making in UK financial services.

Explainable AI and SHAP: Lundberg and Lee (2017) introduced SHAP as a unifying framework for feature attribution, grounded in cooperative game theory, and demonstrated its advantages over earlier methods such as LIME (Ribeiro, Singh and Guestrin, 2016) in terms of theoretical consistency. Subsequent work has applied SHAP specifically to fraud detection contexts; for example, (Bussmann *et al.*, 2021) used SHAP to explain credit risk models and argued that explainability is necessary, not optional, for regulatory acceptance of ML in finance. However, several authors note practical limitations of SHAP at scale, including computational cost for large tree ensembles and the risk that explanations themselves can be misleading if correlated features are not handled carefully (Slack *et al.*, 2020).

Algorithmic fairness: The fairness literature distinguishes between multiple, sometimes mutually incompatible, definitions of fairness, including demographic parity, equalised odds, and individual fairness (Verma and Rubin, 2018; Mehrabi *et al.*, 2022). In financial services specifically, Kozodoi et al. (2022) examined fairness in credit scoring and showed that common fairness metrics can conflict with one another and with overall predictive accuracy, meaning model developers must make explicit, justifiable trade-off decisions rather than assuming fairness is a by-product of good model performance. This trade-off is directly relevant to fraud detection, where false positives (blocking a legitimate customer) and false negatives (missing genuine fraud) carry different costs across different customer groups.

Regulation and Consumer Duty: The FCA's Consumer Duty guidance (Financial Conduct Authority, 2022) sets out cross-cutting rules requiring firms to act in good faith, avoid foreseeable harm, and enable customers to pursue their financial objectives, alongside four outcome rules covering products, price, consumer understanding, and consumer support. While the Duty does not mandate a specific technical method for AI governance, the Bank of England and FCA's joint survey on machine learning in UK financial services (Bank of England, 2022; Financial Conduct Authority, 2022) found that explainability is one of the most frequently cited barriers to greater ML adoption, and that firms are actively seeking practical frameworks to evidence model governance. This positions explainability and fairness auditing not as an academic exercise but as a live, practitioner-relevant problem, though the academic literature has yet to converge on a single applied framework that links technical XAI/fairness output directly to specific Consumer Duty outcomes; this is the gap this dissertation addresses.

This review will be substantially extended in the dissertation literature chapter, incorporating a wider set of sources on model risk management (e.g. SR 11-7 style frameworks as applied in UK practice), additional fairness-aware learning techniques, and recent fraud detection benchmarking studies.

4. Proposed Research Design

This is primarily a quantitative, applied ML research project, structured around building, explaining, and auditing a fraud detection model, then interpreting the results against a regulatory framework.

Data: The project will use the Bank Account Fraud (BAF) Suite (Jesus *et al.*, 2022) as its primary dataset. Published at NeurIPS 2022, BAF comprises six synthetic variants of a real-world bank account opening fraud dataset, generated using differential privacy and a CTGAN-based synthesis process, and was explicitly designed as a benchmark for evaluating both predictive performance and fairness in machine learning. Each variant contains documented data bias across three protected attributes, age group, employment status, and income percentile, removing the proxy-variable uncertainty present in many fraud datasets and providing a direct empirical basis for the fairness audit central to this dissertation. Using public data avoids dependency on gaining access to a real fintech firm's internal system, which the handbook flags as a common feasibility risk, while still being representative of the transaction features (income, device signals, session behaviour, credit risk score) typically used in commercial fraud and account-opening models. The IEEE-CIS Fraud Detection dataset will be used as a secondary validation source, to test whether the explainability and fairness findings generalise beyond a single dataset, addressing a limitation common to single-dataset fraud studies.

Modelling: At least two model types will be trained: an interpretable baseline (logistic regression) and a higher performing but less transparent ensemble model (XGBoost or LightGBM). Standard practice for class-imbalanced fraud data will be followed, including appropriate resampling or class-weighting, and evaluation using precision, recall, F1 and AUC-PR rather than accuracy alone, given the typical rarity of fraud cases.

Explainability: SHAP (via the shap Python library) will be applied to the higher-performing model to generate global feature importance and local, per-prediction explanations. Results will be cross-checked against the logistic regression baseline's coefficients as an interpretability sanity check.

Fairness audit: Group fairness metrics (demographic parity difference, equalised odds difference, disparate impact ratio) will be computed across subgroups using a fairness toolkit such as Fairlearn or AIF360. Results will be analysed for any systematic disparities in false-positive or false-negative rates.

Synthesis: The technical findings (performance, explainability, fairness trade-offs) will then be mapped against the Consumer Duty's outcome rules, using documentary analysis of FCA guidance to assess which technical findings would, and would not, constitute adequate evidence of compliance, and to develop practical recommendations.

5. Project Risk Assessment

As this project uses publicly available data and standard, locally executable ML tooling rather than fieldwork, laboratory work, or third-party data access, overall project risk is low. The principal risks to successful completion are:

- Dataset generalisability risk (low likelihood, low-medium impact): as BAF is synthetically generated, findings may not perfectly transfer to real-world transaction data. Mitigation: the

IEEE-CIS dataset will be used as an independent validation source to test whether explainability and fairness conclusions hold beyond BAF, and any divergence between the two datasets will itself be reported as a finding rather than treated as a failure.

- Computational resource risk (low likelihood, low-medium impact): SHAP computation on large ensembles can be slow. Mitigation: use of TreeSHAP (an efficient SHAP variant for tree-based models), down-sampling of the dataset for SHAP value computation where necessary, and use of university-provided computing resources if local hardware proves insufficient.
- Conflicting fairness metrics risk (medium likelihood, low impact, but important to manage): fairness metrics can disagree with one another, which is a known feature of the field rather than a project failure. Mitigation: this is treated as a substantive finding to be discussed critically, drawing on (Kozodoi, Jacob and Lessmann, 2022), rather than as a problem to be resolved away.
- Scope creep risk (medium likelihood, medium impact): the combination of ML modelling, XAI, fairness auditing and regulatory analysis is broad. Mitigation: the regulatory analysis will be scoped specifically to Consumer Duty (rather than the full UK AI/financial regulation landscape), and the modelling stage will be timeboxed so that sufficient time remains for the explainability and fairness analysis, which is the core contribution.
- Time management risk (low-medium likelihood, medium impact): as with any project, slippage is possible during the modelling or write-up stages. Mitigation: a stage-gated personal timeline will be maintained, agreed against the supervision schedule set out in the PROJ519 handbook, with the pilot model trained and validated before the end of the taught period.

6. Ethics

This project is assessed as low risk from a research ethics perspective and is not expected to require full ethical approval through the Plymouth Ethics Online System (PEOS), for the following reasons: it does not involve human participants, questionnaires, interviews, or primary data collection from individuals; and it uses a secondary, publicly available, pre-anonymised transaction dataset that does not contain personally identifiable information.

Notwithstanding this, two points will be confirmed with the supervisor and, where necessary, raised with the Faculty Research Ethics Committee: first, confirmation that the chosen public dataset's licence and terms of use permit academic research and any redistribution required for marking (e.g. inclusion of code/data access links in the appendix); and second, careful handling of any proxy demographic variables used in the fairness analysis, to ensure that group-level fairness findings are reported and discussed responsibly, without overstating claims about real-world discrimination that the dataset cannot fully support. A formal ethics application will be submitted via PEOS if the supervisor determines, on reviewing the final dataset choice, that one is required.

7. Impact

The project's primary intended impact is practitioner-facing: UK fintech firms and their model risk/compliance teams are actively seeking workable methods for evidencing AI governance under Consumer Duty (Bank of England and FCA, 2022), and this dissertation aims to produce a practical, replicable audit framework rather than a purely theoretical contribution. Potential pathways to impact include sharing the resulting framework and code repository with university careers/industry contacts

in fintech and regtech and adapting the findings into a short practitioner-oriented summary suitable for a compliance or data science audience, distinct from the academic dissertation itself.

The work also has a regulatory and policy-relevant dimension: by identifying where current XAI/fairness techniques fall short of what Consumer Duty implicitly requires, the project can contribute evidence to the ongoing public debate about whether voluntary technical standards are sufficient, or whether more prescriptive regulatory guidance on AI explainability is needed in UK financial services.

In terms of the UN Sustainable Development Goals, the project contributes most directly to SDG 10 (Reduced Inequalities), by addressing the risk of algorithmic systems disproportionately disadvantaging customer groups, and to SDG 9 (Industry, Innovation and Infrastructure), by supporting more trustworthy and accountable deployment of AI infrastructure within the financial sector.

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